

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 11

Probability, Universality, and Levels of Description

Assignment

Self-Study Curriculum

Write a one-page structural analysis of a modeling problem from your own work, organized around levels and universality:

1. Name the level you actually model at (what are the state variables?), and the level(s) below it that you deliberately ignore.
2. What is being coarse-grained away, and why is it safe (or unsafe) to ignore it?
3. Where does concentration of measure earn you the right to use averages? Where might it fail?
4. Is your effective description an instance of universality, would systems with different mechanisms give you the same model? If so, what does that tell you about what your model has and has *not* captured?
5. State honestly which of your model's successes are evidence about the *mechanism* and which are just universality doing the work.